Unit II RADIATION CHEMISTRY

Study of chemical change brought about by high energy radiation is called as radiation chemistry.

Radiolysis: the process in which a substance is exposed to very high energy radiation such as gamma rays is called radiolysis.

Chemical effect due to radiolysis:

The chemical effect due to radiolysis takes place in two stages:

1. Primary stage: in the primary stage of radiolysis the substances absorb high energy radiations and undergo ionization. Ions and electrons are formed.

   $A \xrightarrow{\text{HIGH ENERGY RADIATION}} A^+ + e$

2. Secondary stage: in the secondary stage , ions and electrons give rise to a number of reactions. The reactions are ;

a) Formation of excited molecules

$A^+ + e \longrightarrow A^*$ (excited molecule)

b) Capture of electron by another molecule M to form M^-

$M + e \longrightarrow M^-$

c) Interaction of ions A+ and M- to give excited molecule.

$A+ + M- \longrightarrow M + A^*$ (excited molecule)

d) Decomposition of A* into free radicals X and Y or to give stable molecules B and C.

$A^* \longrightarrow X^{\cdot} + Y^{\cdot}$ (free radicals)

$A^* \longrightarrow B + C$ (stable molecules)

e) Decomposition of ion A+ into another ion R+ and free radicals $X^{\cdot}$ or an ion B+ and a stable compound C.

$A^* \longrightarrow R^+ + X^{\cdot}$

$A^* \longrightarrow B^+ + C$

The free radical formed as above may initiate other chain reactions.

Radiolysis of water vapour:

Radiolysis of water vapour by gamma radiations gives following major products.

a) Formation of ionic products:

The following ionic products are formed.

$H_2O + Y \longrightarrow H_2O^*$ (excited) $\longrightarrow H_2O^+$

$H_2O + e \longrightarrow H^+ + OH^{\cdot} + 2e \longrightarrow H_2O^+ + 2e$

$H_2O + e \longrightarrow OH^+ + H^{\cdot} + 2e \longrightarrow H_2O^+ + 2e$

$H_2O + e \longrightarrow O^+ + H_2 + 2e$

$H_2O^+ + H_2O \longrightarrow H_3O^+ + OH^{\cdot}$

$H_2O^+ + 2e \longrightarrow e\,(aq)$

$H_2O + Y \longrightarrow H_2O^*$ (excited) $\longrightarrow H_2O^+$

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$H_2O + e \longrightarrow O^+ + H_2 + 2e$

$H_2O^+ + H_2O \longrightarrow H_3O^+ + OH^{\cdot}$

$H_2O^+ + 2e \longrightarrow e\,(aq)$

The electrons formed either recombine with positive ions or get solvated as shown above. Experimently it has been found that H_2O^+ is the main ionic product of radiolysis of water vapour.

b) Formation of free radicals: the ionic products formed as above are very reactive. They recombine to form free radicals.

$H_2O^+ + e \longrightarrow OH^{\cdot} + H^{\cdot}$

$H^+ + H_2O + e \longrightarrow H_3O^+ + e \longrightarrow 2\,H^{\cdot} + OH^{\cdot}$

$OH^+ + H_2O + e \longrightarrow H^{\cdot} + 2OH^{\cdot}$

$H^{\cdot} + O_2$ (from dissolved air or O_2) $\longrightarrow HO_2^{\cdot}$ (Perhydroxyl)

c) Formation of molecular products:

The free radicals formed above are highly reactive and combine among themselves to form stable molecular products.

$H^{\cdot} + OH^{\cdot} \longrightarrow H_2O$

$H^{\cdot} + H^{\cdot} \longrightarrow H_2$

$OH^{\cdot} + OH^{\cdot} \longrightarrow H_2O_2$

$OH^{\cdot} + OH_2^{\cdot} \longrightarrow H_2O + O_2$

$OH_2^{\cdot} + OH_2^{\cdot} \longrightarrow H_2O_2 + O_2$

d) Reaction between molecules and free radicals:

Under favourable experimental conditions, reactions between free radicals and molecules take place leading to chain reactions.

$H^{\cdot} + H_2O_2 \longrightarrow H_2O + OH^{\cdot}$

$OH^{\cdot} + H_2 \longrightarrow H^{\cdot} + H_2O$

e) Formation of hydrated / solvated electrons:

The electrons formed above get hydrated producing an coloured hydrated ion, e^- (aq) and is highly reactive. It readily reacts with water and H^+ ions. The reaction is

$H_2O + e^-\,(aq) \longrightarrow H^{\cdot} + OH^-$

$H^+ + e^-\,(aq) \longrightarrow H$

f) Scavenging the free radicals:

In the radiolysis of water, the secondary reactions are due to free radicals as ionic products are short lived and molecular products are inert.

The secondary reaction can be eliminated by removing free radicals by adding a suitable substance in trace amounts which is capable of removing the free radicals. The substance is called scavenger and the process is called scavenging. Deuterium scavenges both $H^{\cdot}$ and $OH^{\cdot}$ radicals.

$D_2 + H^{\cdot} \longrightarrow D^{\cdot} + HD$

$D_2 + OH^{\cdot} \longrightarrow D^{\cdot} + HOD$

$D^{\cdot} + D^{\cdot} \longrightarrow D_2$ (at the walls of the container)

Radiolysis of benzene:

Radiolysis of benzene results in the formation of biphenyl and hydrogen. The possible reaction is

$C_6H_6 + Y \longrightarrow C_6H_6^+ + e$ primary process

$C_6H_6^+ + e \longrightarrow C_6H_6^*$

$C_6H_6^* \longrightarrow C_6H_5 + H^\cdot$

$C_6H_5 + C_6H_5 \longrightarrow C_6H_5 - C_6H_5$ (Biphenyl)

Radiolysis of Acetic acid:

Radiolysis of acetic acid results in the formation of succinic acid and hydrogen. The possible reaction is

$CH_3COOH + Y \longrightarrow CH_3COOH^+ + e$ primary process

$CH_3COOH^+ + e \longrightarrow CH_3COOH^*$

$CH_3COOH^* \longrightarrow CH_2COOH + H^\cdot$

$CH_2COOH + CH_2COOH \longrightarrow$ CH_2COOH succinic acid

CH_2COOH

$H^\cdot + H^\cdot \longrightarrow H_2$

<u>Theories of Radiolysis:</u>

The important theories are, **1) Linde's Cluster theory**

2) Eyring, Hirchelder and Taylor (EHT) theory

1) Linde's Cluster theory:

This theory explains large ionic yields in radiation reactions. We are aware that the primary product of radiation is ions. Around these ions gather many neutral molecules to form a cluster. The neutral molecules are held by induced polarization forces. When the cluster is neutralized by electrons , the energy released makes all the molecules in the cluster to under go reaction or changes.

Example:

Synthesis of NH_3: When a mixture of N_2 and H_2 is irradiated high energy radiation the following reaction take place.

$N_2 + \gamma$ (radiation) $\longrightarrow N_2^+ + e$ ……..(primary process)

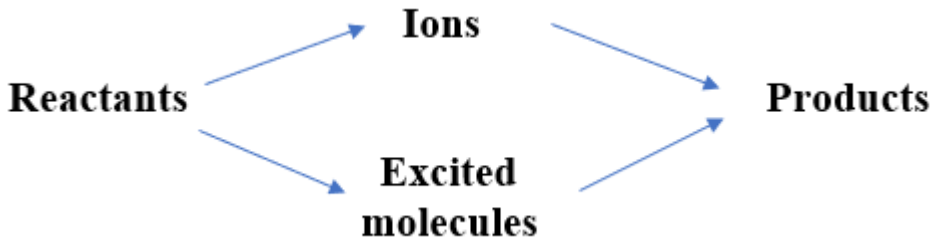
$N_2^+ + 3H_2 \longrightarrow N_2H_6^+$ ……..Cluster

$N_2H_6^+ + e \longrightarrow 2NH_3$

2) Eyring, Hirchelder and Taylor (EHT) Theory:

According to this theory a part of the radiation energy is utilized in ionization and another part is in producing the excited molecules. Both the ions and the excited molecules give the products.

The EHT theory is summarized as,



Radiation dosimeter units:

The amount of radiation energy absorbed is called Dose. The dose is expressed in different units. They are,

1. **Rad(r) :** The absorption of 100 ergs of energy per gram of the substance is called rad.
   1 rad $=10^{-5}$ Jg^{-1} = 6.24 × 10^{13} eVg^{-1}
2. **Gray(Gy) :** This is a SI unit to express the dose. It is defined as the absorption of one joule of radiation energy per kilogram of matter.
   1 gray = $1Jkg^{-1}$ = 100 rad
3. **Roentgen (R) :** It is defined as that quantity of dose that would produce in one cm^3 of dry air atSTP, ions carrying 1 esu of charge of either sign.
   1 roentgen = 0.87 rad = 8.7×10^3 Gy
4. **Ionic Yield : T**he number of molecules that undergo reaction per 100 eV of energy absorbed is called ionic yield.

$$\text{Ionic Yield} = \frac{No\ of\ molecules\ reacted}{no\ of\ ions\ pairs\ formed\ in\ the\ same\ time}$$

5. **Energy yield or Radiation yield or G- value:**

The number of molecules reacted or produced for each 100 eV of energy absorbed is called energy yield.

Dosimeter:

The instrument used to measure the quantity of radiation absorbed by substance is called dosimeter.

The dosimeter consists of (i) A substance whose chemical change due to radiation serves as a measure of absorbed energy (ii) A glass vessel to irradiate the substance.

Example:

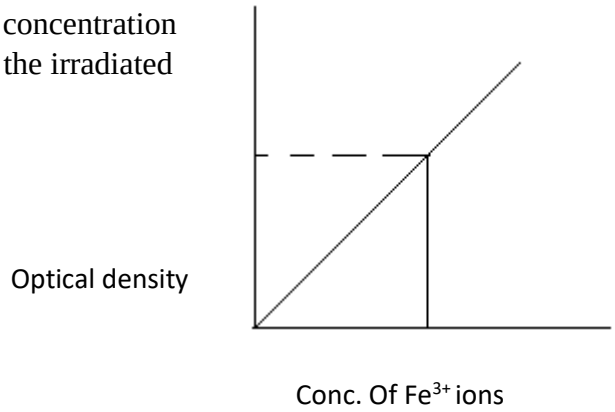
Fricke Dosimeter:

Based on the study of the oxidation of Fe^{2+} to Fe^{3+} ions. The dosimeter solution is prepared by dissolving 0.4 g of Mohr's salt, 0.06 g of NaCl and 22 cm^3 of concentrated H_2SO_4 in 1 dm^3 of distilled water. Analytical grade chemicals and distilled water used for radiation experiments are to be used.

A part of this solution is taken in a thick glass container and exposed to γ- radaiation for a certain period of time. Fe^{2+} is oxidized to Fe^{3+} ions. NaCl present facilitates the oxidation reaction.

The optical densities for different concentrations of Fe^{3+} ions are determined using a spectrophotometer at a wavelength of 3040 Å. The optical density of the irradiated sample is

also determined. A graph of optical density versus concentration is plotted. From the graph concentration of Fe^{3+} in the irradiated sample can be determined.



Knowing that the standard yield of Fe^{3+} ions is 15.5 ions per 100 eV of energy absorbed, the amount of radiation energy absorbed in eV during the irradiation can be calculated.

The reaction mechanism is as follows;

$H_2O \xrightarrow{\gamma \text{ radiation}} H^{\cdot}, OH^{\cdot}, HO^{\cdot}_2, H_2, H_2O_2$

$Fe^{2+} + OH^{\cdot} \longrightarrow Fe^{3+} + OH^{-}$

$Fe^{2+} + HO^{\cdot}_2 \longrightarrow Fe^{3+} + OH_2^{-}$

$HO^{-}_2 + H^{+}(H_2SO_4) \longrightarrow H_2O_2$

$Fe^{2+} + H_2O_2 \longrightarrow Fe^{3+} + OH^{\cdot} + OH^{-}$

Difference between Radiation chemistry and photochemistry:

Radiation Chemistry	Photochemistry
1.energy if the quantum is low.	1.High energy quantum is used usually greater than 100 eV
2.Only solute undergo reaction.	Solute, solvent and impurities also react.
3.Actinometers are used to measure the absorbed radiation.	3.Dosimeters are used to measure the absorbed radiation.
4.Quantum yield is calculated	4. Ionic yield is calculated.